CCBBA
EACCD
DDADD
CDDCA
EEADA
BDAB

3.6 firm entry and exit

1. A profit-maximizing firm will shut down in the short run any time the firm's total revenue is less than its

(A) total cost
(B) fixed cost
(C) total variable cost ✓
(D) explicit cost
(E) implicit cost

2. Assume that a profit-maximizing, perfectly competitive firm has economic losses in the short run. If the firm continues to produce and sell its goods, then which of the following must be true?

(A) The firm is covering all of its fixed and variable costs of production.
(B) The firm is covering all of its fixed costs but not all of its variable costs of production.
(C) The firm is covering all of its variable costs but not all of its fixed costs of production. ✓
(D) The firm is covering all of its implicit costs but not all of its explicit costs.
(E) The firm must have raised the price of its goods in order to minimize its losses.

3. Assume that, for a perfectly competitive firm, marginal cost equals average variable cost at $10, marginal cost equals average total cost at $15, and marginal revenue equals marginal cost at $12. On the basis of this information, the firm should

(A) close down in the short run
(B) operate in the short run, even though it will sustain a loss ✓
(C) operate in the short run, because it will make an economic profit of $3 per unit
(D) operate in the long run, because it will make an economic profit of $3 per unit
(E) operate in the short run, but decrease output to decrease its cost

4. Assume that olive oil is produced in a constant- cost, perfectly competitive industry, which is currently in long-run equilibrium. If the current price of olive oil is $5 per quart and the demand for olive oil increases, then the price of olive oil will change in which of the following ways in the short run and long run?

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(A)

Short Run	Long Run
Be more than $5	Be more than $5

(B) ✓

Short Run	Long Run
Be more than $5	Be equal to $5

(C)

Short Run	Long Run
Be equal to $5	Be equal to $5

(D)

Short Run	Long Run
Be less than $5	Be less than $5

(E)

Short Run	Long Run
Be less than $5	Be equal to $5

Assume a perfectly competitive firm is currently producing 100 units of output. Its marginal cost is $\$6$ and rising at that output quantity. Its average variable cost is $\$7$ and its average fixed cost is $\$3$.

5. If the product's price is $\$6$, which of the following will the firm do in the short run to maximize its profit?

(A) Shut down ✓

(B) Produce, but less than 100 units of output

(C) Produce more than 100 units of output

(D) Continue to produce at exactly 100 units of output

(E) Increase its price above $\$6$

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Answer A

Correct. The firm currently produces the 100 units where marginal cost equals marginal revenue. The price, $6, is less than the average variable cost. By continuing to operate, the firm incurs a loss greater than its fixed cost. Therefore, the firm will shut down in the short run to minimize its losses.

6.

Daily Output	Total Variable Cost
0	$0
1	20
2	30
3	48
4	72
5	100

Based on the cost and output data in the table above, a perfectly competitive firm will shut down if price falls below

(A) $30
(B) $20
(C) $18
(D) $16
(E) $15 ✓

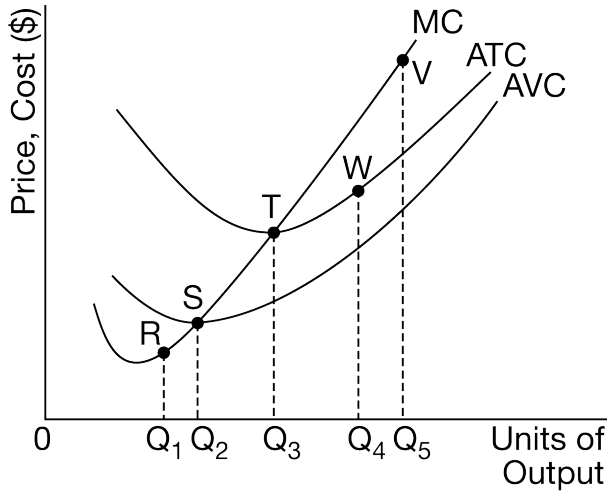
7. In the absence of barriers to entry, a typical firm is currently in long-run equilibrium. Assume there is an increase in the market demand for the good that the firm is producing. Which of the following will happen in the long run?

(A) New firms will enter the market. ✓
(B) The market supply will decrease, but the quantity supplied will increase.
(C) The firm will earn positive economic profit.
(D) The firm's price will be greater than its average revenue.
(E) The firm will continue to produce the same quantity of output.

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Answer A

Correct. The increase in market demand shifts the market demand curve to the right, resulting in an increase in the price. As a result, all existing firms will increase output to where marginal cost equals marginal revenue and earn positive economic profit in the short run. In the long-run, incentivized by the short-run profits, new firms will enter the market, driving economic profits down to zero.



The graph above shows the cost curves for May's Fruit Farm, where MC is marginal cost, ATC is average total cost, and AVC is average variable cost.

8. May's short-run supply curve includes which of the following points?

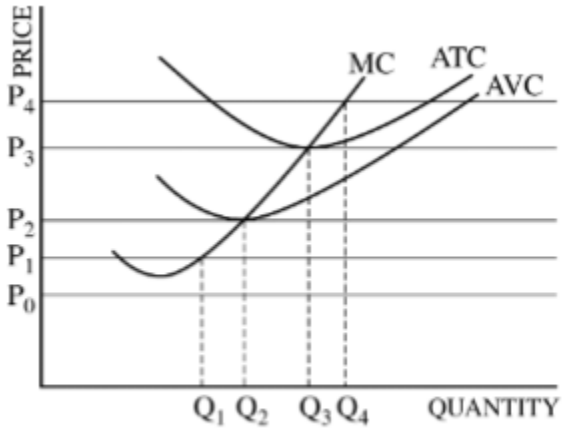
(A) TW
(B) RST
(C) STV ✓
(D) STW
(E) RSTV

Answer C

Correct. A firm's (May's) short-run supply curve is the portion of the marginal cost curve that lies above the minimum AVC. On the graph that is line segment STV.

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Question is based on the following graph, which shows a firm's marginal cost (MC), average total cost (ATC), and average variable cost (AVC).



9. In the short run, the firm will

(A) shut down if the price falls below P_3

(B) continue to produce as long as the price is greater than P_1

(C) continue to produce as long as the price is greater than P_2 ✓

(D) earn economic profits as long as the price is greater than P_2

(E) cover its fixed cost as long as the price is less than P_3

10. The firm's short-run supply curve is which of the following?

(A) The AVC curve above P_1

(B) The AVC curve above P_2

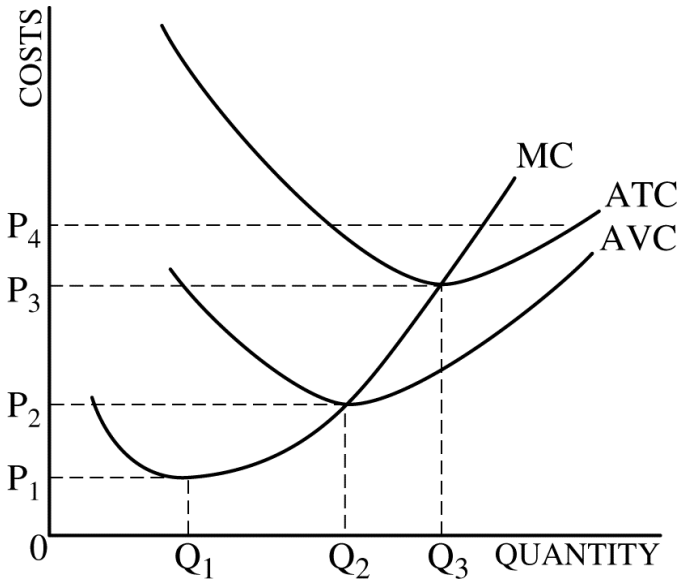
(C) The ATC curve above P_3

(D) The MC curve above P_2 ✓

(E) The MC curve above P_3

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The following problems refer to the graph below for a representative firm in a perfectly competitive, constant-cost industry, which shows the firm's marginal cost (MC), average total cost (ATC), and average variable cost (AVC).



11. In the short run, the firm will realize an economic loss but will continue to produce if the price is

(A) below P_1

(B) between P_1 and P_2

(C) below P_2

(D) between P_2 and P_3 ✓

(E) between P_3 and P_4

12. Which of the following MUST be true in the long run?

(A) The equilibrium price will be P_2, since that is where marginal cost equals minimum average variable cost.

(B) The equilibrium price will be above P_3, since firms must make an economic profit to stay in business.

(C) If the price is above P_2, new firms will enter the industry.

(D) If the price is above P_3, new firms will enter the industry. ✓

(E) If the price is above P_4, firms will exit the industry.

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13. For a perfectly competitive, increasing-cost industry, an increase in the industry's demand will lead to which of the following in the long run?

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(A) An upward shift in each firm's long-run average cost curve ✓

(B) An increase in each firm's profit

(C) A decrease in the price of an input and a decrease in total industry profits

(D) A decrease in total industry sales

(E) A decrease in total producer surplus and an increase in total consumer surplus

14. In order to minimize short-run losses, a profit-maximizing firm will necessarily shut down production under which of the following conditions?

(A) Total revenue is less than total cost.

(B) Marginal cost is greater than average total cost.

(C) Marginal cost is less than marginal revenue.

(D) Average revenue is less than average variable cost. ✓

(E) Average revenue is less than average total cost.

15. In the short run, a profit-maximizing firm should shut down if which of the following is true?

(A) It is not making an economic profit.

(B) It is not making a normal profit.

(C) Its total revenue is less than its total cost.

(D) Its product price is less than its average variable cost. ✓

(E) Its product price is greater than its average variable cost but less than its average total cost.

16. In the short run, if a firm produces the level of output at which marginal revenue is equal to marginal cost but price is less than average total cost, the firm will

(A) always shut down production

(B) expand output to lower its average fixed cost

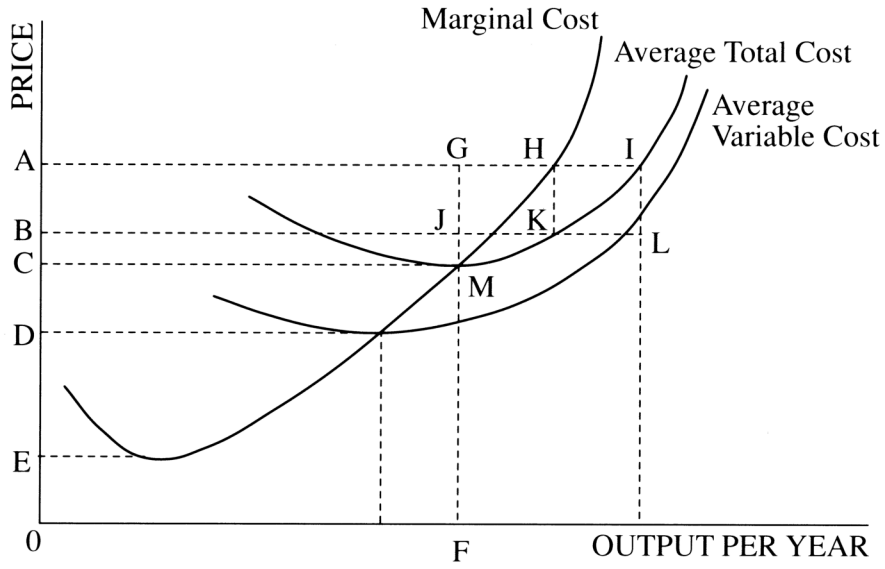
(C) continue to operate if price is greater than its average variable cost ✓

(D) decrease output until price equals its average total cost

(E) increase output to increase revenue

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Refer to the following diagram and assume a perfectly competitive market structure.



17. In the short run, the firm will stop production when the price falls below

(A) 0A

(B) 0B

(C) 0C

(D) 0D ✓

(E) 0E

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The following questions are based on the table below, which shows a firm's average variable cost and average total cost.

Output	Average Variable Cost	Average Total Cost
1	$40	$160
2	35	95
3	40	80
4	45	75
5	50	74
6	55	75

18. In the short run, the lowest price at which the firm will continue to produce is

(A) $80
(B) $74
(C) $50
(D) $35 ✓
(E) $30

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19. JC pizzeria has a year remaining on an unbreakable lease on its building, requiring a payment of $20,000 a year. If JC operates over the next year, it estimates that its revenues will be $200,000 and that its expenses, in addition to the lease, will be $190,000. Which of the following statements is true?

(A) JC should shut down, since it will incur a loss of $20,000.
(B) JC should shut down to break even.
(C) JC should operate, since its loss is less than its fixed cost. ✓
(D) JC should operate, since it will earn a profit of $10,000.
(E) JC will break even, whether it operates or shuts down.

20. If a typical firm in a perfectly competitive market earns positive economic profit in the short run, what will most likely happen in the long run?

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(A) Firms will enter the market and cause the price to fall. ✓
(B) Firms will enter the market and cause the price to rise.
(C) Firms will exit the market and cause the price to fall.
(D) Firms will exit the market and the cause price to rise.
(E) There will be no change in the market because profit is maximized.

Answer A

Correct. The positive economic profit attracts new firms to enter the market, increasing supply and putting a downward pressure on price. This will eliminate economic profits in the long run.

21. The short-run supply curve for a firm in a perfectly competitive industry is

(A) its entire marginal cost curve
(B) its average variable cost curve above its marginal cost curve
(C) its average total cost curve above its marginal cost curve
(D) its marginal cost curve above the minimum point of its average total cost curve
(E) its marginal cost curve above the minimum point of its average variable cost curve ✓

22. OfficeClips is a typical firm that produces and sells paper clips in a perfectly competitive market. OfficeClips is currently earning a positive economic profit. In the long run, what will happen to the firm's economic profit, the market price, and the market quantity?

	Firm's Profit	Market Price	Market Quantity
A	No change	Decrease	Increase
B	No change	Increase	Decrease
C	Decrease	Increase	Decrease
D	Decrease	Decrease	Decrease
E	Decrease	Decrease	Increase

(A) A
(B) B
(C) C
(D) D
(E) E ✓

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Answer E

Correct. When a typical firm is earning positive economic profit in a perfectly competitive market, new firms will enter the market. As firms enter the market, market supply will increase, causing the market price to decrease and the market quantity to increase. This results in a decrease in the firm's profit.

23. In the short run, a firm in a perfectly competitive industry will definitely

(A) shut down if total revenue is less than total variable cost ✓

(B) produce the market equilibrium output to maximize profit

(C) earn zero economic profit

(D) cover all of its fixed costs

(E) charge a price lower than its competitors to increase its sales

Answer A

Correct. In the short run, a perfectly competitive firm will continue to produce when total revenue exceeds total variable cost (or the price is greater than average variable cost). The firm will shut down when the firm's total revenue is less than its total variable cost to minimize the losses, which will be equal to the fixed costs in the short run.

24. In the short run, which of the following must be true for a perfectly competitive firm that is maximizing profits?

(A) The firm will shut down if it has any economic losses.

(B) The firm will produce at the minimum of average total cost.

(C) The firm will produce where $\mathrm{MR} = \mathrm{MC}$, but price from the demand curve is greater than MC.

(D) The firm will produce where $\mathrm{MR} = \mathrm{MC}$ as long as P is greater than average variable cost. ✓

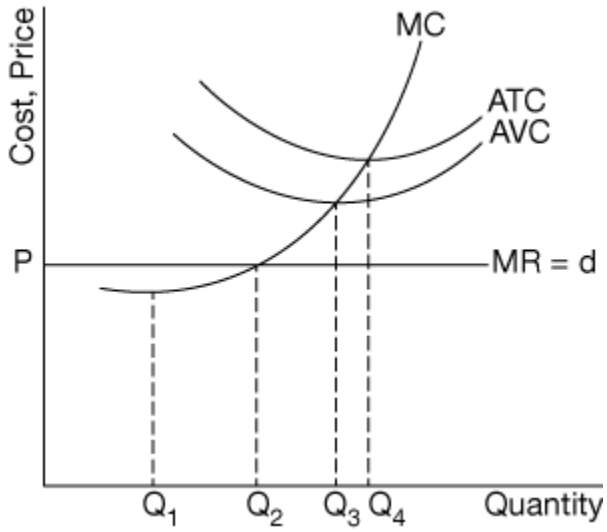
(E) The firm will produce the quantity that exhibits allocative and productive efficiency.

Answer D

Correct. The profit-maximizing firm will produce where $\mathrm{MR} = \mathrm{MC}$. If price is greater than ATC, the firm has economic profits. The firm will still produce with economic losses as long as P is greater than average variable cost, as the losses will be less than fixed costs, which would be the losses if the firm shuts down.

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25.



The graph provided shows the demand (d), marginal revenue (MR), marginal cost (MC), and average total cost (ATC) curves for a perfectly competitive firm. In order to maximize short-run profits, this perfectly competitive firm should produce at which of the following quantities of production?

(A) Zero ✓

(B) Q_1

(C) Q_2

(D) Q_3

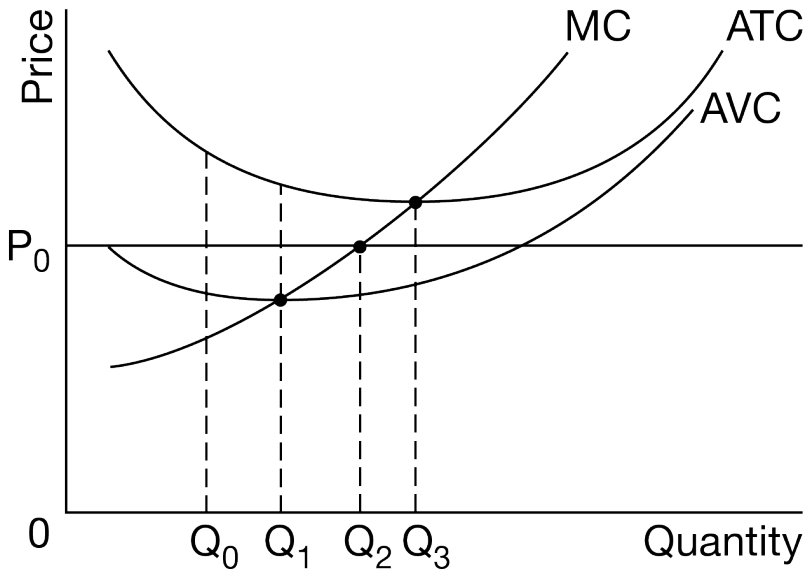
(E) Q_4

Answer A

Correct. The firm should shut down (produce zero units of output) because the price is less than the average variable cost throughout the range of production.

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26.



The graph above shows the short-run cost curves for a perfectly competitive firm. Assume that the market price is P_0 and the firm is producing at quantity Q_0. To maximize profit, the firm should

(A) increase production to quantity Q_3, where average total cost is at its minimum

(B) increase production to quantity Q_2, where price is equal to marginal cost ✓

(C) increase production to quantity Q_1, where average variable cost is at its minimum

(D) shut down and incur losses equal to fixed costs

(E) shut down and incur zero losses

Answer B

Correct. The profit-maximizing perfectly competitive firm should produce where P (or MR) $= MC$, at Q_2. The firm will have a loss, since P is less than ATC. The firm should continue to produce, since $P > AVC$ and the loss will be less than fixed costs.

27. At its current level of output, a firm's total revenue is greater than its total variable cost but less than its total cost. If the firm is producing at the point where marginal revenue is equal to marginal cost, what should the firm do to maximize profit in the short run?

(A) Increase price to increase revenue.

(B) Decrease price to increase quantity.

(C) Exit the market to minimize losses.

(D) Continue to produce at its current level of output to minimize losses. ✓

(E) Shut down.

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Answer D

Correct. The firm will minimize its losses by continuing to operate in the short run. Its total revenue is greater than its total variable cost. Because it earns more than enough to cover its total variable costs, the excess amount over the total variable cost covers part of its fixed cost. Thus the firm minimizes its losses by continuing to operate rather shutting down.

28. A profit-maximizing firm is currently producing a quantity at which price is less than average variable cost. To maximize profit, the firm will do which of the following in the short run and the long run?

(A) Shut down in the short run and exit the market in the long run. ✓

(B) Produce less quantity in the short run and increase its scale of production in the long run.

(C) Produce more quantity in the short run and increase its scale of production in the long run.

(D) Continue producing the same quantity in the short run and in the long run.

(E) Continue producing the same quantity in the short and exit the market in the long run.

Answer A

Correct. The firm is profit-maximizing, therefore it is producing the quantity at which MR equals MC. However, when price is less than average variable cost, the firm is better off shutting down to minimize its costs to fixed costs only in the short run and exit the market in the long run. By continuing to produce in the short run, the firm is unable to cover its variable costs and part of its fixed costs because price is less than average variable cost.

29. In the short run, when a profit-maximizing, perfectly competitive firm is producing at the quantity where $\text{MR} = \text{MC}$ and the firm's total revenue is less than its total variable cost, the firm should

(A) continue to produce the same quantity

(B) shut down ✓

(C) lower its price

(D) raise its price

(E) expect other firms to enter the industry

Answer B

Correct. A firm must shut down in the short run if its total revenue is less than its variable cost, as the

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firm would no longer be able to pay for costs that vary with output.